

John Hoeksema
Software Engineer

CONTACT

(616) 238-9895 < > johnmarchhoeksema@gmail.com < > [LinkedIn](#) < > [github](#) < > [personal website](#)

EXPERIENCE

Software Requirements Engineer

Gentex

April 2024 - January 2026

- Developed greenfield state machine requirements for the next gen Video Processing software
- Developed software requirements for 4 LCD Mirror product lines used in over 60,000 vehicles annually
- Lead the Video Processing team and managed product scope for 2 weeks in the Project Manager's absence

Software Analyst

Accenture Federal Services

August 2021 - September 2022

- Scripted mass emailing to decrease time cost for the Department of Veterans Affairs Digital GI Bill project
- Performed data analysis and implemented Decision Tree machine learning on veteran data

NLP Developer, Intern

X2AI

Spring 2020

- Trained BERT neural net to detect emotional valence in 4000+ text messages with 90+% accuracy
- Structured and tested data preprocessing pipeline that expanded contractions, eliminated filler words, and used lemmatization and tokenization

Drone Researcher

Department of Computer Science, University of Notre Dame

Summer 2019

- Developed code for drone fleet emergency response system for use by the South Bend Fire Department
- Implemented basic computer communication with network configurations and routing algorithms

PROJECTS

Brain Pong

Summer 2026

- Built a real-time [EOG](#) pipeline using detrend, bandpass filtering, σ calibration, and velocity thresholding on an ADS1299 board to move a Pong paddle in under 500 ms
- Designed a state machine that rejects involuntary saccades; benchmarked detectors at 94% accuracy and 400 ms median latency
- Curated a 237-recording, 23-subject EOG corpus with pseudonymization and a [public web viewer](#)

Explore Mars

Spring 2026

- Built a 3D Mars globe with 20 GB of terrain imagery, CloudFront + S3, and 2,000+ searchable features
- Optimized render performance by batching draw calls and reducing rover datapoints from 50,000 to 500
- Engineered mobile-first UI/UX with custom touch controls and progressive terrain loading

EDUCATION

B.S. in Computer Science, University of Notre Dame

May 2021

SKILLS

Proficient: Agentic coding, TypeScript, React, Python, Git, Data Preprocessing

Familiar: C/C++, PyTorch, SQL, Linux, Go, Raspberry Pi/Arduino, Jira